

Sihua Zhang 张思华

Undergraduate Student, School of Artificial Intelligence, Southern University of Science and Technology (SUSTech)
12410106@mail.sustech.edu.cn | github.com/dove667 | dove667.github.io

EDUCATION

Southern University of Science and Technology (SUSTech) 2024 – 2028 (expected)

B.Eng. in Artificial Intelligence, School of Artificial Intelligence

- GPA: 3.75 / 4.0 (Year 1), 3.94 / 4.0 (Year 2)
- Research Interests: Natural Language Processing · Multimodal Large Language Models · Full-Duplex Omni Models · LLM Optimization Methods

RESEARCH EXPERIENCE

Research Assistant — Guanhua Chen's Group, Department of Statistics and Data Science 2026 – Present

Southern University of Science and Technology (SUSTech)

- Working on full-duplex Omni models that listen, think, and speak simultaneously — spoken-dialogue systems that interject and stay in lock-step with real human conversation.
- Current focus: low-latency speech-text streaming, real-time interruption handling, and tightly coupling audio and text generation inside one unified model.

PROJECTS

Muon Optimizer on nanoGPT [GitHub](#) · [Report](#)

Course project, Mathematical Foundations of AI — SUSTech

- Studied how Newton-Schulz orthogonalization schedules in the Muon optimizer affect GPT pretraining (100M tokens, 1× RTX 4090, FineWeb-10B).
- Muon variants reach final validation loss 4.28–4.53 vs 5.37 for AdamW; multi-seed runs confirm fast coefficients (fast5) as the strongest practical baseline.
- Showed that deeper orthogonalization (manual / Polar Express) improves update-matrix semi-orthogonality, especially on attention projections, without further loss gains.

What Determines Long-Context Ability in LLMs? — Survey with Experimental Analysis [GitHub](#) · [Report](#)

Course project, NLP — SUSTech

- Decomposed long-context ability into nominal window, effective retrieval length, long-range reasoning, and system efficiency; benchmarked a dense Transformer (Llama-3.1-8B-Instruct) against an SSM (Falcon3-Mamba-7B-Instruct) on RULER and LongBench.
- Profiled HF vs vLLM serving (TTFT / TPOT / peak memory) for long-context workloads.

SUSTech RAG Q&A System [GitHub](#)

Course project — SUSTech

- Built a RAG-based Q&A system covering the full pipeline from data acquisition to retrieval and generation, including frontend and backend services.

SKILLS

Languages

Python C / C++ CUDA

Frameworks & Tools

PyTorch HuggingFace vLLM Scikit-learn
OpenCV Git Linux Docker

SELECTED SELF-STUDY COURSEWORK

6.S184 Generative AI with SDEs — Flow & Diffusion Models (MIT, IAP 2026) · CSE 234: Data Systems for Machine Learning (UC San Diego) · CS 285: Deep Reinforcement Learning (UC Berkeley) · CS336: Building Language Models From Scratch (Stanford) · Advanced NLP (CMU) · CS236: Deep Generative Models (Stanford) · Dive into Deep Learning